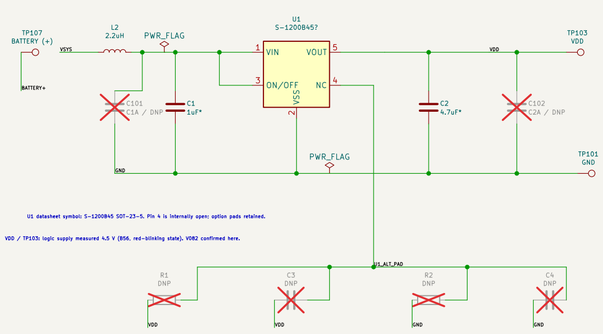
## 01 POWER / INPUT FILTER

U1: S-1200B45 candidate; VDD measured 4.5 V. Device identity remains qualified.



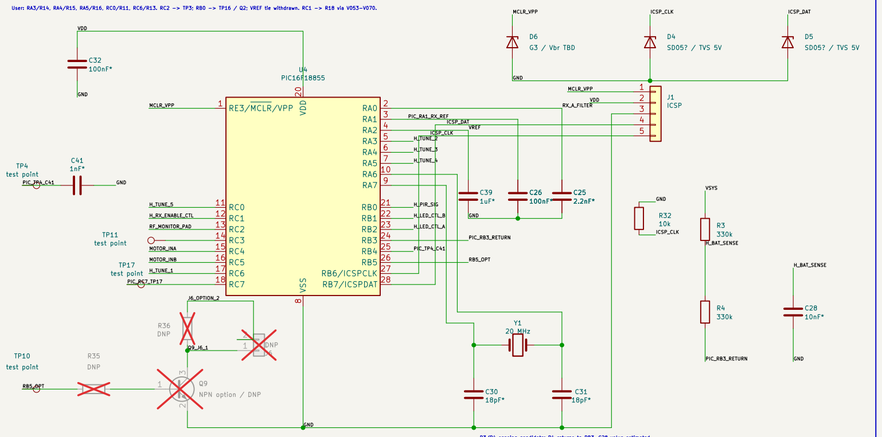

U1 datasheet symbol: S-1200B45 SOT-23-5. Pin 4 is internally open; option pads retained.

VDD / TP103: logic supply measured 4.5 V (B56, red-blinking state). VDB2 confirmed here.

Empty options: R1/C3 free ends join regulated VDD; R2/C4 free ends join GND.

## 02 PIC CONTROLLER / CLOCK / ICSP

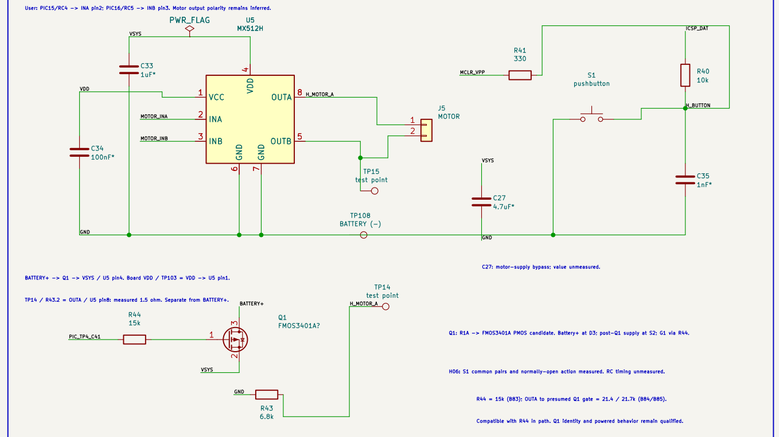
User: RA3/R14, RA4/R15, RA5/R16, RC0/R11, RC6/R13, RC2 -> TP3; RB0 -> TP16 / Q2; VREF tie withdrawn, RC1 -> R18 via V053-V070.




R3/R4 sensing candidate: R4 returns to RB3. C28 value estimated.

## 03 MOTOR / BUTTON / BATTERY INTERFACE

User: PIC15/RC4 -> INA pin2; PIC16/RC5 -> INB pin3. Motor output polarity remains inferred.




C27: motor-supply bypass; value unmeasured.

BATTERY+ -> Q1 -> VSYS / U5 pin4. Board VDD / TP103 -> VDD -> U5 pin1.

TP14 / R43.2 = OUTA / U5 pin8: measured 1.5 ohm. Separate from BATTERY+.




Q1: R1A -> FMOS3401A PMOS candidate. Battery+ at D3; post-Q1 supply at S2; G1 via R44.

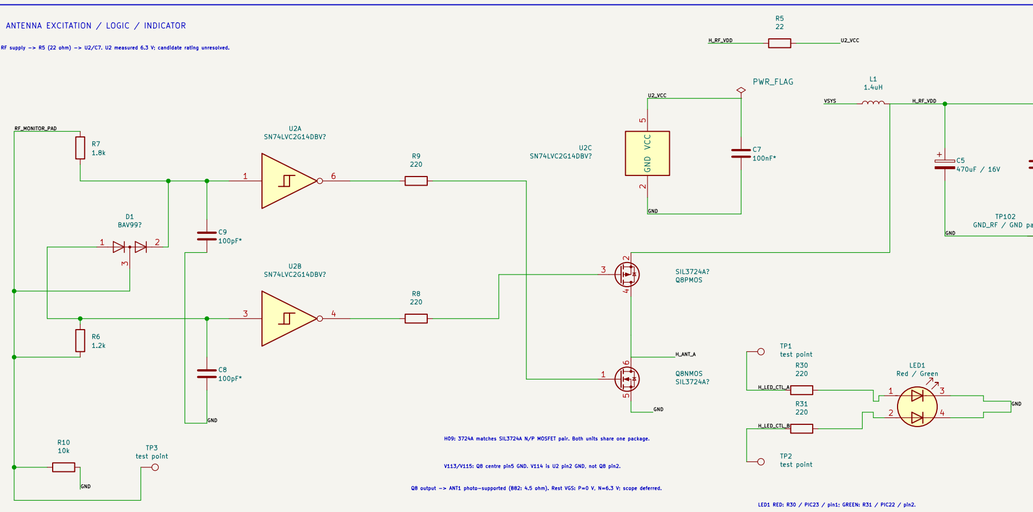
H06: S1 common pairs and normally-open action measured. RC timing unmeasured.

R44 = 15k (B83); OUTA to presumed Q1 gate = 21.4 / 21.7k (884/885).

Compatible with R44 in path. Q1 identity and powered behavior remain qualified.

## 04 ANTENNA EXCITATION / LOGIC / INDICATOR

H07: RF supply -> R5 (22 ohm) -> U2/C7. U2 measured 6.3 V; candidate rating unresolved.




H09: 3724A matches SIL3724A N/P MOSFET pair. Both units share one package.

V113/V115: Q8 centre pin5 GND. V114 is U2 pin2 GND, not Q8 pin2.

Q8 output -> ANT1 photo-supported (B82: 4.5 ohm). Rest VGS: P=0 V, N=6.3 V; scope deferred.

LED1 RED: R30 / PIC25 / pin1; GREEN: R31 / PIC22 / pin2.

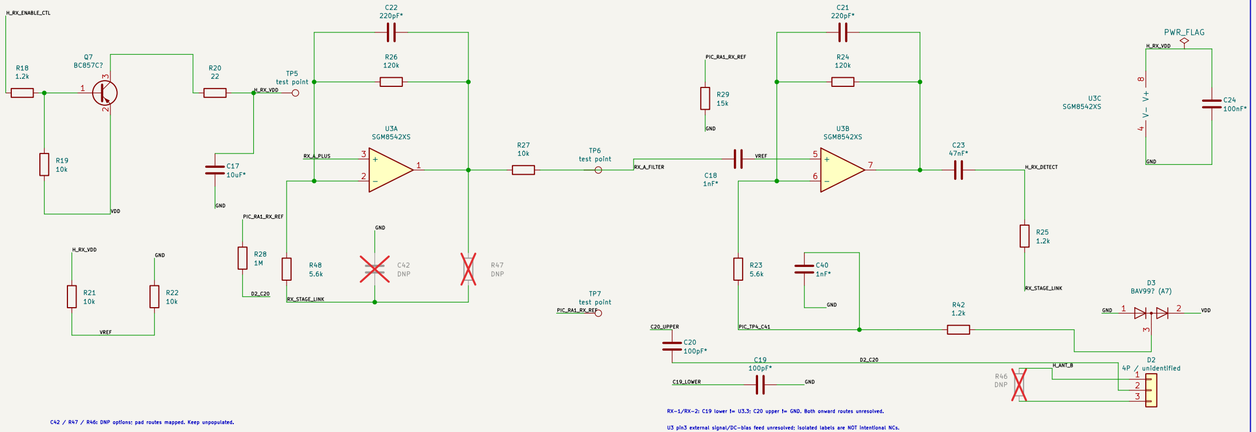
IMG_2431: TP3/PIC13 drives R6/R7/D1 midpoint. Diode identity and timing unverified.

## 05 RF RECEIVE / GAIN / ENVELOPE

H10–H12: working topology around photographed 120k feedback resistors. Filters, bias, and interstage routes inferred.

Q7: 11.2 kohm V064–V070 ffs R18+R19. Supply-switch topology inferred: RC1 drives R18 via V053–V070.

User: D3.R -> V082 / VDD; D3.L -> GND: D3.S -> R42 1.2 ohm (photo selects pad2).




C42 / R47 / R46: DNP options: pad routes mapped. Keep unpopulated.

U3 alternative: MCP6002-I/SN: same pin roles. electrical suitability to verify.

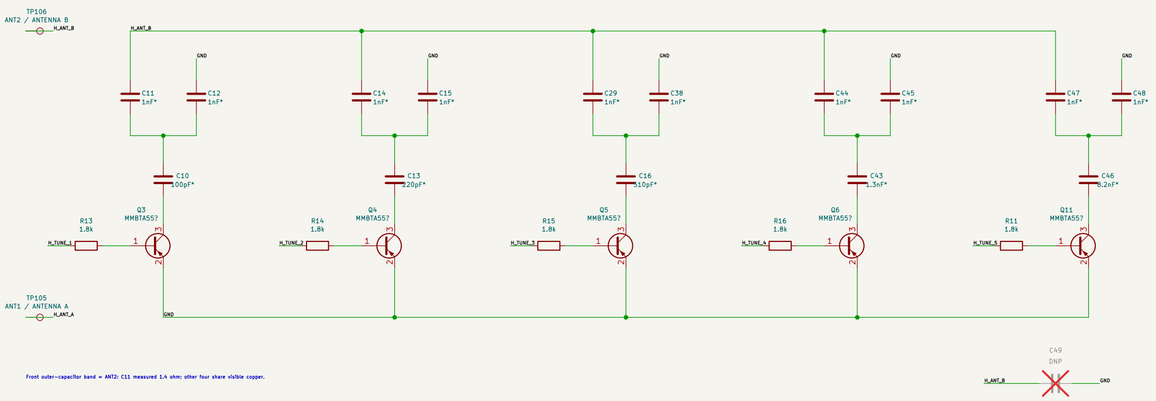
User: U3 pin5 / V079 / V080 = VREF (1 ohm): TP6 remains separate across C18.

RX-1/RX-2: C19 lower I= U3.3; C20 upper I= GND. Both onward routes unresolved.

U3 pin3 external signal/DC-bias feed unresolved: isolated labels are NOT intentional NCs.

## 06 SWITCHED ANTENNA CAPACITANCE

H13: 2H device identity/pinout disputed. Starred values are a 500-uH RF design seed.



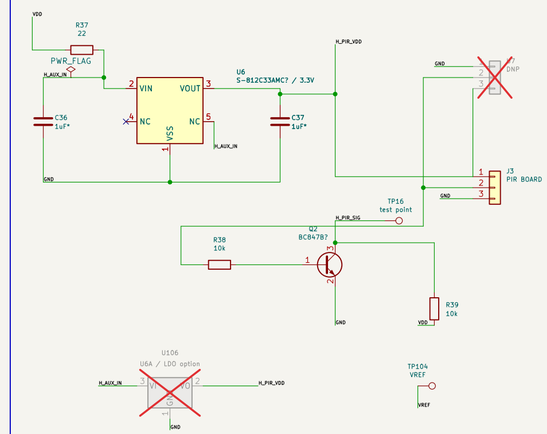

Front outer-capacitor bend = ANT2: C11 measured 1.4 ohm; other four share visible copper.

Rear capacitor returns remain GND. Front and rear capacitors are not drawn in parallel.

Q3 diode pattern disputes PNP pinout; 2H family and RF switching role remain unverified.

## 07 PIR / AUXILIARY RAIL

U6: C2NM / S-812C33 candidate; PIR supply measured 3.3 V. Input follows R37.




U6: pin5 joins VIN (1 ohm); pin4 unused. PIR supply measured 3.3 V with sensor attached.

U6A: DNP alternative regulator: stock symbol does not identify the original part.

## READING THIS WORKING RECONSTRUCTION

One sheet. Functional wiring is proposed: this is not a verified production circuit.

148 retained catalog entries: unsupported TP9 and R33 entries withdrawn.

H_* = inferred inter-block net. Other net names do not imply measurement.

* = proposed capacitor value, not measured. ? = candidate device identity or pin role.

Library pin numbers are mapped to photo pads in evidence/pin_crosswalk.json.

U2 and U3 use multiple symbol units on this same sheet, including supply units.

28 original local photo fragments retained: Q1.L–R43.2 withdrawn by user.

RA0 reaches TP6. TP11/TP17 local routes retained; hidden branches unproved.

DNP = unpopulated. An open pin means unresolved, not a certified no-connect.

PIR daughterboard internals are not yet traced; J3 represents its interface.

C5: 470uF/16V marked. DC rails measured: 44 starred ceramic values are estimates.

See docs/CAPACITOR_VALUE_REVIEW.html for values, ratings and assumptions.

Antenna: 500 uH reported: RF seed 134.2 kHz. Ideal switches: weights/order unverified.

Proposed voltage, dielectric, tolerance and evidence are in capacitor properties.

v0.9.47 | KiCad 10.0.5 | 2026-10-15

44 starred capacitor estimates; antenna reported 500uH; RF weights unverified.

**KiCad 10 libraries + four datasheet symbols | MCP authoring**

Sheet: /

File: PetSafe_1001339.kicad_sch

**Title: PetSafe 100–1339 R03 A | single-sheet reconstruction**

| Size: A2 | Date: 2026-10-15 | **Rev: 0.9.47** |
|---|---|---|
| KiCad E.D.A. 10.0.5 | | Id: 1/1 |